Beam Me Up

Light travels in straight lines until it hits something. Then it can bounce, bend, soak in, pass through, or spread out, and your eye is built to catch it.

My name: _____

My partners: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 23	Bounce the Beam
Lesson 2	Fri Dec 4	Angle In Equals Angle Out
Lesson 3	Fri Jan 22	Full Challenge Practice
Lesson 4	Fri Feb 19	Mock Tournament
Lab I1	Before Lesson 2	Five Things Light Can Do (home or an open Friday)
Lab I2	Before Lesson 3	Inside the Eye (home or an open Friday)
Lab I3	Before Lesson 4	Mirror Path Planner and Test Review (home or an open Friday)

At the tournament

Teams of 3 take a 50-minute written test on light and the eye, and are called out one team at a time to a darkened room to bounce a fixed flashlight beam off 3 handheld mirrors plus 1 wall mirror onto a target, twice, in 60 seconds or less each (bring only pencils and 1 protractor).

Keep every handout for this event behind this divider, in order. Bring this binder to every Build & Test Fridays meeting.

Beam Me Up: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can explain the two parts of Beam Me Up (written test and 4-mirror challenge) and the rules for the challenge.	L1 L3 L4	☐	☐
I can tilt a mirror in small, steady moves to put a flashlight spot on a target.	L1 L2 L3	☐	☐
I can define reflection and name things that reflect light well (mirrors, foil, still water).	L1 L3 I1	☐	☐
I can state the law of reflection: the angle in equals the angle out, both measured from the normal line.	L2 L3 I3	☐	☐
I can use a protractor to measure the angle of a light beam hitting a mirror.	L2 L4 I3	☐	☐
I can draw and predict the path of light bouncing off one or more mirrors.	L2 L3 I3	☐	☐
I can define refraction and explain why a pencil looks bent in a glass of water.	L3 I1	☐	☐
I can define absorption and explain why dark clothes get warmer in the sun.	L3 I1	☐	☐
I can define transmission and sort materials as transparent, translucent, or opaque.	L3 I1	☐	☐
I can define scattering and use it to explain why the sky looks blue and why a beam shows up in fog or dust.	L3 I1	☐	☐
I can label all 11 eye parts: ciliary muscle, conjunctiva, cornea, iris, lens, optic nerve, photoreceptors (rods and cones), pupil, retina, sclera, vitreous humor.	L4 I2	☐	☐
I can explain the job of each eye part and trace light from the cornea to the brain.	L4 I2	☐	☐
I can tell rods from cones and explain why colors are hard to see in dim light.	L4 I2	☐	☐
I can plan a mirror path in 1 minute and call clear directions as Team Leader or mirror holder.	L3 L4 I3	☐	☐
I can finish a challenge in under 60 seconds by holding the beam on the target for 3 seconds.	L3 L4	☐	☐

Bounce the Beam

TODAY I CAN

- I can explain the two parts of Beam Me Up (written test and 4-mirror challenge) and the rules for the challenge.
- I can tilt a mirror in small, steady moves to put a flashlight spot on a target.
- I can define reflection and name things that reflect light well (mirrors, foil, still water).

Welcome to Beam Me Up! Today you will steer a beam of light with mirrors, just like the real challenge.

Today's steps: run it yourselves (22 minutes)

You need: 4 handheld mirrors, stopwatch, painter's tape, flashlight on its stand (already set up; do not move it). Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read 'The event in a nutshell' out loud. Everyone points to the flashlight, the wall mirror, and the two targets.
2. Team 5 min. Set the timer for 5 minutes. Ask the adult in the room before you dim the lights. Each person catches the beam with 1 mirror and moves the spot onto the big target. Use two hands, tiny turns, and never shine light at anyone's eyes.
3. Team 6 min. Set the timer for 6 minutes. Partner A catches the beam and passes it to Partner B, who passes it to Partner C, who aims at the target. Switch jobs once. Fill rows 1 to 3 of 'Our tries'.
4. Event Judge 5 min. Set the timer for 5 minutes. Add the wall mirror. The team gets 1 minute with lights on to plan. Lights off, count 3-2-1, start the stopwatch, and stop when the spot stays on the target 3 seconds (or at 60). Write row 4.
5. Team 3 min. Lights on. Set the timer for 3 minutes. Answer 'Think about it', then check with the answer card.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Compare your 'Think about it' answers to the good answers on the card and add one idea you missed in a different color.

The event in a nutshell

Team	3 students. One is the Team Leader, who tells the judge when you are ready.
Written test	Questions about light and the parts of the eye.
Challenge	Use 3 handheld mirrors and 1 wall mirror to send a flashlight beam onto a target.
Time	1 minute to plan with the lights on. Up to 60 seconds in the dark. The beam must stay on the target for 3 seconds.
Reflection	When light bounces off something. Mirrors reflect light very well.

Our tries

Try	Mirrors used	Target (big or small)	Time (seconds)	Hit it? (yes/no)
1	1			
2	2			
3	3			
4	4 (with wall mirror)			

Think about it

1. What happened to the spot on the wall when you tilted your mirror just a little bit?

2. What is one thing your team did that helped you hit the target faster?

At home before next time: Do Independent Lab I1 (Five Things Light Can Do) before Lesson 2. It starts with a Light Hunt at home. Then make 8 flashcards: reflection, refraction, absorption, transmission, scattering, transparent, translucent, opaque (word on the front, meaning and an example on the back). Engineer's Logbook: build the Mirror Maze (p. 13) for the Beam Bouncer badge, and plan your first mirror path on graph paper (p. 5).

Angle In Equals Angle Out

TODAY I CAN

- I can tilt a mirror in small, steady moves to put a flashlight spot on a target.
- I can state the law of reflection: the angle in equals the angle out, both measured from the normal line.
- I can use a protractor to measure the angle of a light beam hitting a mirror.
- I can draw and predict the path of light bouncing off one or more mirrors.

Mirrors follow one simple rule. If you know the rule, you can plan your path before the lights go out.

Today's steps: run it yourselves (22 minutes)

You need: 2 slit flashlights (index card taped over the lens), 2 handheld mirrors and poster putty, 2 paper protractors, 3 more handheld mirrors, 1 big target and painter's tape, stopwatch. Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read 'Words to know' out loud. Partner A draws a mirror, a dashed normal line, and a beam in and out on the whiteboard. The team checks that both angles are measured from the normal line.
2. Team 7 min. Set the timer for 7 minutes. Lay the handout flat and stand a mirror on the 'Ray lab' line with putty. Shine the slit beam at the mirror and trace the beam in and out. Measure both angles from the normal and fill 'Ray lab data' for 3 different angles.
3. Team 3 min. Set the timer for 3 minutes. Go to the floor protractor under the wall mirror. Partner B stands at 40 degrees on the left. Everyone points to where Partner C should stand to catch the bounce (40 degrees on the right).
4. Event Judge 6 min. Set the timer for 6 minutes. Tape the target somewhere new. The team uses the floor protractor to plan, then runs 1 timed try with all 4 mirrors from outside the taped box. Ask the adult in the room before you dim the lights.
5. Team 3 min. Lights on. Set the timer for 3 minutes. Answer 'Predict', then check with the answer card.

Check your work: Ask your Event Captain for the Lesson 2 answer card. Your angle in and angle out should be within about 5 degrees of each other. Check both 'Predict' answers and fix any you missed in a different color.

Words to know

Normal line	An imaginary line that sticks straight out from the mirror (a square corner, 90 degrees).
Angle of incidence	The angle between the beam coming in and the normal line.
Angle of reflection	The angle between the beam going out and the normal line.
Law of reflection	The angle in always equals the angle out.

Ray lab: stand your mirror on this line

_____ mirror _____

Ray lab data

Try	Angle in (degrees)	Angle out (degrees)	Equal? (yes/no)
1			
2			
3			

Predict

1. A beam hits a mirror at 35 degrees from the normal. What is the angle of reflection?

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2. On the floor protractor, the beam comes into the wall mirror from 50 degrees on the left. Where should the next mirror holder stand?

At home before next time: Do Independent Lab I2 (Inside the Eye) before Lesson 3. Then make 11 eye flashcards (part on the front, job on the back). Keep quizzing your 8 light-word flashcards. Engineer's Logbook: build a Periscope (p. 14) and explain its 45 degree angles for the Periscope Pro badge.

Full Challenge Practice

TODAY I CAN

- I can explain the two parts of Beam Me Up (written test and 4-mirror challenge) and the rules for the challenge.
- I can tilt a mirror in small, steady moves to put a flashlight spot on a target.
- I can define reflection and name things that reflect light well (mirrors, foil, still water).
- I can state the law of reflection: the angle in equals the angle out, both measured from the normal line.
- I can draw and predict the path of light bouncing off one or more mirrors.
- I can define refraction and explain why a pencil looks bent in a glass of water.
- I can define absorption and explain why dark clothes get warmer in the sun.
- I can define transmission and sort materials as transparent, translucent, or opaque.
- I can define scattering and use it to explain why the sky looks blue and why a beam shows up in fog or dust.
- I can plan a mirror path in 1 minute and call clear directions as Team Leader or mirror holder.
- I can finish a challenge in under 60 seconds by holding the beam on the target for 3 seconds.

Today you run the full challenge, just like at the tournament. Plan, talk, and hold still.

Today's steps: run it yourselves (27 minutes)

You need: 3 handheld mirrors, 3 stopwatches, flashlight on its stand (already aimed; do not touch). Set the timer for each step.

1. Team 3 min. Set the timer for 3 minutes. Read 'Challenge strategy' out loud. Pick the Team Leader, then decide the beam's order around the furniture and where each mirror holder stands.
2. Event Judge 8 min. Red course (big target). Set the timer for 8 minutes. 1 minute to plan with lights on; the Team Leader says Ready. Ask the adult in the room before you dim the lights. Wait 10 seconds, 3-2-1, go, and write the middle of the 3 times in 'Our times'. Run it twice if you can.
3. Event Judge 8 min. Blue course (small target). Set the timer for 8 minutes. Same routine. After each try, the team says one thing it could change.
4. Team 5 min. Lights on. Set the timer for 5 minutes. Answer 'Light review' together, then check with the answer card.
5. Team 3 min. Set the timer for 3 minutes. Add the Red and Blue times, divide by 2, and fill the last row of 'Our times'. Say one thing that will make you faster.

Check your work: Ask your Event Captain for the Lesson 3 answer card and check 'Light review'. Fix any you missed in a different color. For each challenge try, check yourself against the rules: did the beam hit all 4 mirrors and stay on the target 3 full seconds?

Challenge strategy

1. Lights on: find the flashlight, the wall mirror, the target, and anything in the way.
2. Decide the order the beam will travel and where each mirror holder stands (outside the taped box).
3. Use angle in equals angle out at the wall mirror to find where to stand.
4. Team Leader says Ready. In the dark, Mirror 1 finds the beam and freezes. Then Mirror 2, then Mirror 3.
5. Tiny moves only. Hold the spot on the target for 3 seconds.

Our times

Course	Team Leader	Plan time (sec)	Challenge time (sec)
Red (big target)			
Blue (small target)			
Average of the two			

Light review

1. A straw in a glass of water looks bent. This is:

◯ Reflection ◯ Refraction ◯ Absorption

2. A black car seat gets hot in the sun because the seat:

◯ Reflects light ◯ Transmits light ◯ Absorbs light

3. You can see car headlight beams in fog because of:

◯ Scattering ◯ Refraction ◯ Transmission

4. Wax paper lets some blurry light through. It is:

◯ Transparent ◯ Translucent ◯ Opaque

5. Seeing yourself in a still pond is:

◯ Absorption ◯ Reflection ◯ Scattering

6. Light hits a mirror at 30 degrees from the normal. It leaves at:

◯ 60 degrees ◯ 90 degrees ◯ The same angle, 30 degrees, on the other side of the normal

At home before next time: Do Independent Lab I3 (Mirror Path Planner and Test Review) before Lesson 4. It has you quiz all 19 flashcards (5 light words, 3 see-through words, 11 eye parts). Write the 3 cards you missed most on the back of this page. Engineer's Logbook: build a Pinhole Eye (p. 15) and see the upside-down picture, like your retina, for the Eye Engineer badge.

Mock Tournament

TODAY I CAN

- I can explain the two parts of Beam Me Up (written test and 4-mirror challenge) and the rules for the challenge.
- I can use a protractor to measure the angle of a light beam hitting a mirror.
- I can label all 11 eye parts: ciliary muscle, conjunctiva, cornea, iris, lens, optic nerve, photoreceptors (rods and cones), pupil, retina, sclera, vitreous humor.
- I can explain the job of each eye part and trace light from the cornea to the brain.
- I can tell rods from cones and explain why colors are hard to see in dim light.
- I can plan a mirror path in 1 minute and call clear directions as Team Leader or mirror holder.
- I can finish a challenge in under 60 seconds by holding the beam on the target for 3 seconds.

Practice test. Stop when you are called for the challenge, then come back and finish.

Today's steps: run it yourselves (27 minutes)

You need: 3 handheld mirrors, 3 stopwatches, 1 protractor per team, flashlight on its stand (already set; do not touch), Lesson 4 answer card (Event Captain keeps it face down). Set the timer for each step.

1. Event Judge 2 min. Set the timer for 2 minutes. Read the handout intro out loud. Teams write their names on the test.
2. Team 15 min. Set the timer for 15 minutes. Start the practice test. The Event Judge calls one team at a time to the challenge corner for two courses (1 minute to plan, lights off, 3-2-1, 60 seconds max) and writes the middle times in 'Score box'. Then the team goes back to the test.
3. Event Captain 6 min. Set the timer for 6 minutes. Read each answer from the Lesson 4 answer card. Teams mark their own tests and talk about any question they missed.
4. Team 4 min. Set the timer for 4 minutes. Average the two challenge times and fill 'Score box'. Write one thing you will do the same way at the tournament.

Check your work: The Event Captain reads the Lesson 4 answer card while teams mark their own tests. Circle each missed question and find the right answer on your Independent Lab I1 or I2 page.

Light

1. Light passing through a clear window is:
○ Transmission ○ Reflection ○ Scattering

2. Light that turns into heat when it hits dark pavement is:
○ Refraction ○ Absorption ○ Transmission

3. Light bending as it goes from air into water is:
○ Reflection ○ Refraction ○ Scattering

4. The sky looks blue because air bits:
○ Absorb red light ○ Scatter blue light the most ○ Reflect like a mirror

5. A beam hits a mirror at 40 degrees from the normal. The angle of reflection is:
○ 20 degrees ○ 40 degrees ○ 80 degrees

The eye

1. The colored muscle ring that controls the pupil:
○ Iris ○ Retina ○ Lens

2. The clear front dome that does most of the light bending:
○ Sclera ○ Cornea ○ Conjunctiva

3. Carries signals from the eye to the brain:
○ Optic nerve ○ Ciliary muscle ○ Vitreous humor

4. Photoreceptors that work best in dim light:
○ Cones ○ Rods ○ Pupils

5. Clear jelly that keeps the eye round:

◯ Vitreous humor ◯ Lens ◯ Cornea

6. Changes the shape of the lens to focus:

◯ Iris ◯ Ciliary muscle ◯ Optic nerve

7. Thin clear skin over the white of the eye and inside the eyelids:

◯ Conjunctiva ◯ Retina ◯ Sclera

8. Tough white outer coat of the eye:

◯ Pupil ◯ Sclera ◯ Lens

Explain

1. Put light's path in order: lens, optic nerve, cornea, retina, pupil, vitreous humor.

2. What is strange about the picture on the retina, and what fixes it?

3. Which of the five light words does a mirror show best? Why?

Score box

Part	Our result
Test score (out of 16)	
Red course time (sec)	
Blue course time (sec)	
Average challenge time (sec)	

At home before next time: Tournament prep: redo every question you missed today using your flashcards and your Independent Lab I2 eye page. Pack 2 pencils and 1 protractor. Decide who your Team Leader will be and practice calling directions with two family members and a flashlight if you can. Engineer's Logbook: run the Living-room 4-Mirror Course (p. 16) as Team Leader for the Beam Captain badge, and use the Tournament-day checklist (p. 17).

Lab I1: Five Things Light Can Do

THIS LAB HELPS ME

- I can define reflection and name things that reflect light well (mirrors, foil, still water).
- I can define refraction and explain why a pencil looks bent in a glass of water.
- I can define absorption and explain why dark clothes get warmer in the sun.
- I can define transmission and sort materials as transparent, translucent, or opaque.
- I can define scattering and use it to explain why the sky looks blue and why a beam shows up in fog or dust.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. When light hits something, one of five things can happen. You can do this lab alone or with your team, at home or at the Open Lab table on a Friday your group does not meet.

Steps (28 minutes)

You need: flashlight, Station A: foil, white paper, black paper, Station B: plastic wrap, wax paper, cardboard, Station C: 2 clear cups, pencil, coin, water, Station D: clear cup, spoon, milk or powdered creamer, paper towels. Set the timer for each step.

1. Me 4 min. Set the timer for 4 minutes. Light Hunt: walk around with a flashlight. Find 5 things that reflect light well, 3 things light passes through, and 3 things that block light. Fill the 'Light Hunt' table.
2. Me 3 min. Set the timer for 3 minutes. Read 'The five light words' out loud. After each word, point to one thing from your Light Hunt that shows it.
3. Me 4 min. Station A. Set the timer for 4 minutes. In a dim spot, shine the flashlight on foil, then white paper, then black paper, aimed so the bounce lands on a wall. Which sends the most light to the wall? Write it in 'Station notes'.
4. Me 4 min. Station B. Set the timer for 4 minutes. Shine the light through plastic wrap, wax paper, and cardboard onto the table. Write transparent, translucent, or opaque for each in 'Station notes'.
5. Me 5 min. Station C. Set the timer for 5 minutes. Put the pencil in a cup of water and look from the side. Coin trick: put the coin in the empty cup, lower your head until the rim just hides the coin, keep still, and slowly pour in water. Write what you saw.
6. Me 4 min. Station D. Set the timer for 4 minutes. Stir a little milk or creamer into a cup of water (a little cloudy, not white). Shine the light through the side. Look from the side, then from the far end, and write the colors.
7. Me 4 min. Set the timer for 4 minutes. Do 'Match each example to a light word' and answer 'Think about it'. Then check with the I1 answer card.

Check your work: Get the I1 answer card from the Open Lab bin (or ask your Event Captain at Lesson 2). Check your Station notes, matching, and 'Think about it'. Fix any you missed in a different color.

Light Hunt

Reflects light well (shiny)	Lets light through	Blocks light

The five light words

Reflection	Light bounces off a surface. Smooth, shiny things reflect best.
Refraction	Light bends when it goes from one material into another, like air into water.
Absorption	Light soaks into a material and turns into heat. Dark colors absorb the most.
Transmission	Light passes through a material.
Scattering	Light bounces off tiny bits (dust, water drops, air) in many directions. Air scatters blue light most, so the sky looks blue.

Transparent / Translucent / Opaque	Clear, light passes through / Some blurry light passes through / No light passes through.

Station notes

Station	What I did	What I saw	Light word
A	Light on foil, white, black paper		
B	Light through wrap, wax paper, cardboard		
C	Pencil in water; coin trick		
D	Light through cloudy water		

Match each example to a light word

Sunglasses	Transmission
Pencil in water	Scattering
Flashlight beam in fog	Reflection
Window	Absorption
Mirror	Refraction

Think about it

1. Why does a black shirt feel warmer in the sun than a white shirt?

2. Why did the coin appear when you poured in water?

Bring to the next meeting: Bring your Station notes and your 8 light-word flashcards to Lesson 2.

Lab I2: Inside the Eye

THIS LAB HELPS ME

- I can label all 11 eye parts: ciliary muscle, conjunctiva, cornea, iris, lens, optic nerve, photoreceptors (rods and cones), pupil, retina, sclera, vitreous humor.
- I can explain the job of each eye part and trace light from the cornea to the brain.
- I can tell rods from cones and explain why colors are hard to see in dim light.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Your eye catches light and sends signals to your brain. Learn these 11 parts for the test. Do this lab alone or with a teammate.

Steps (29 minutes)

You need: small mirror, hand lens and white index card, colored pencils, eye worksheet (optional). Set the timer for each step.

1. Me 4 min. Set the timer for 4 minutes. Pupil watch: look at one eye in the mirror. Cover it with your hand for 10 seconds, then uncover it and watch the black dot. Draw it before and after in 'Pupil watch'. Never look at the sun or shine a light into your eye.
2. Me 8 min. Set the timer for 8 minutes. Read 'The 11 eye parts' one row at a time, in the order light travels. After each row, add that part to your side-view eye in the draw box. Label all 11.
3. Me 5 min. Set the timer for 5 minutes. Model eye: stand away from a bright window (never aim at the sun). Hold the hand lens toward the window and move the white card behind it until a small, sharp picture appears. What do you notice?
4. Me 4 min. Set the timer for 4 minutes. In a dim closet or dark room, pull 4 colored pencils out without looking and name each color. Check in bright light. Read the rods and cones rows to explain what happened.
5. Me 5 min. Set the timer for 5 minutes. Answer 'Check yourself'.
6. Me 3 min. Set the timer for 3 minutes. Cover the facts box. Point to each part on your drawing and say its job. Then check with the I2 answer card.

Check your work: Get the I2 answer card from the Open Lab bin (or ask your Event Captain at Lesson 3). Check 'Check yourself', then check your 11 labels against the facts box. Fix any you missed in a different color.

The 11 eye parts (keep this page!)

Conjunctiva	Thin, clear skin that covers the white of the eye and the inside of the eyelids. Keeps the eye moist and protected.
Cornea	Clear dome on the front of the eye. Light enters here. It does most of the bending (refraction) of light.
Sclera	The tough white outer coat. Protects the eye and keeps its shape.
Iris	The colored ring. It is a muscle that makes the pupil bigger or smaller.
Pupil	The black hole in the middle of the iris. Lets light into the eye.
Lens	Clear, bendy part behind the pupil. Changes shape to focus light on the retina.
Ciliary muscle	Ring of muscle around the lens. Pulls to change the lens shape so we can focus near or far.
Vitreous humor	Clear jelly that fills the inside of the eye and keeps it round. Light passes through it.
Retina	Thin layer at the back of the eye where the picture lands (upside down).
Photoreceptors: rods	Light-catching cells in the retina. Work in dim light. See black, white, and gray.
Photoreceptors: cones	Light-catching cells in the retina. See color and fine detail. Need bright light.
Optic nerve	Carries signals from the retina to the brain. The brain makes the picture and flips it right side up.

Pupil watch: draw your eye before (covered) and after (in the light)

Before After

Draw and label a side view of the eye (all 11 parts)

Check yourself

1. Put in order: retina, cornea, optic nerve, lens, pupil, vitreous humor.

2. What did the pupil do when the light got bright? Which part made it happen?

3. In the model eye, what was the picture on the card like?

4. Why is it hard to see colors in a dark room?

5. What does the sclera do?

Bring to the next meeting: Bring your labeled eye drawing and your 11 eye flashcards to Lesson 3.

Lab I3: Mirror Path Planner and Test Review

THIS LAB HELPS ME

- I can state the law of reflection: the angle in equals the angle out, both measured from the normal line.
- I can use a protractor to measure the angle of a light beam hitting a mirror.
- I can draw and predict the path of light bouncing off one or more mirrors.
- I can plan a mirror path in 1 minute and call clear directions as Team Leader or mirror holder.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. At the tournament you get only 1 minute to plan your mirror path. This lab trains your angle brain and your memory. Do it alone or with teammates.

Steps (28 minutes)

You need: protractor, ruler, your 19 flashcards, small mirror and flashlight (optional). Set the timer for each step.

1. Me 3 min. Set the timer for 3 minutes. Read 'Mirror rules' out loud. Say the law of reflection without looking.
2. Me 6 min. Set the timer for 6 minutes. In 'Protractor practice', draw 3 mirrors with a dashed normal line on each. Use the protractor to draw a beam coming in at 20, 45, and 70 degrees, then draw each beam going out at the same angle.
3. Me 5 min. Set the timer for 5 minutes. Answer 'Angle problems'. Draw a quick sketch for any that are hard.
4. Me 6 min. Set the timer for 6 minutes. Solve the 'Path puzzle'. Draw where each mirror goes and the beam's path. If you have a mirror and flashlight, try one bounce for real.
5. Both 3 min. Set the timer for 3 minutes. Write 5 short calls a Team Leader could use in 'Caller words'. Say them out loud to a teammate or family member. Alone is fine too.
6. Me 5 min. Set the timer for 5 minutes. Shuffle all 19 flashcards and quiz yourself. Tally right and missed in 'Flashcard check'. Then check your answers with the I3 answer card.

Check your work: Get the I3 answer card from the Open Lab bin (or ask your Event Captain at Lesson 4). Check 'Angle problems' and hold your protractor on each drawing: angle in and angle out must match. Fix any you missed in a different color.

Mirror rules

Normal line	An imaginary line straight out from the mirror, at 90 degrees.
Law of reflection	Angle in = angle out, both measured from the normal line.
Mirror surface trick	If you know the angle from the mirror's surface, subtract it from 90 to get the angle from the normal.
Straight on	A beam that comes in along the normal (0 degrees) bounces straight back.

Protractor practice: 3 mirrors, beams in at 20, 45, and 70 degrees

Angle problems

1. A beam hits a mirror 15 degrees from the normal. What is the angle of reflection?

2. A beam leaves a mirror 55 degrees from the normal. What was the angle of incidence?

3. A beam hits a mirror 30 degrees from the mirror's surface. What is its angle from the normal?

4. The beam in and the beam out make an 80 degree angle with each other. What is the angle of incidence?

5. A beam hits a mirror straight on, along the normal. Where does it go?

6. On the floor protractor, the beam comes into the wall mirror from 25 degrees on the left. Where should the next mirror holder stand?

Path puzzle

A flashlight sits in the bottom-left corner of a room, pointing right. The wall mirror is on the right wall. The target is on the top-left wall. Draw the room, 3 handheld mirrors, and the wall mirror so the beam hits all 4 mirrors, then the target.

Caller words

Call	What it means

Flashcard check

Cards	Right	Missed	Cards to study again
5 light words			
3 see-through words			
11 eye parts			

Bring to the next meeting: Bring this planner and your 3 most-missed flashcards to Lesson 4.